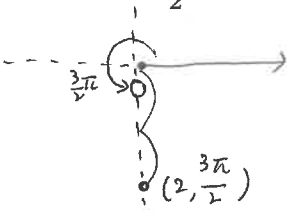
2026년 1학기 수학1 3차 손과제					
학과		학번		이름	

§10.3 Polar Coordinates (극좌표)

1. Plot the point whose polar coordinates are given. Then find the Cartesian coordinates of the point.

(1) $(2, \frac{3\pi}{2})$

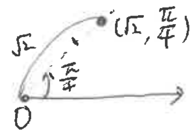


$x = 2\cos\frac{3}{2}\pi = 0$

$y = 2\sin\frac{3}{2}\pi = -2$

∴ $(0, -2)$ ///

(2) $(\sqrt{2}, \frac{\pi}{4})$

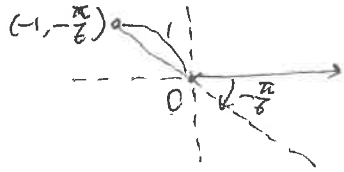


$x = \sqrt{2}\cos\frac{\pi}{4} = 1$

$y = \sqrt{2}\sin\frac{\pi}{4} = 1$

∴ $(1, 1)$ ///

(3) $(-1, -\frac{\pi}{6})$



$x = (-1)\cos(-\frac{\pi}{6}) = (-1)\frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{2}$

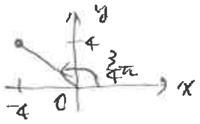
$y = (-1)\sin(-\frac{\pi}{6}) = (-1)\cdot(-\frac{1}{2}) = \frac{1}{2}$

∴ $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$ ///

2. The Cartesian coordinates of a point are given. Find polar coordinates (r, θ) of the point, where **(a)** $r > 0$ and $-\pi < \theta \le \pi$, and **(b)** $r < 0$ and $0 \le \theta < 2\pi$.

(1) $(-4, 4)$

x y



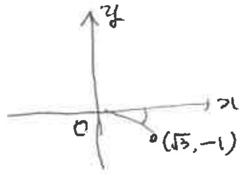
$r^2 = x^2 + y^2 = 32$

$\tan\theta = \frac{4}{-4} = -1$

(a) $(4\sqrt{2}, \frac{3}{4}\pi)$

(b) $(-4\sqrt{2}, \frac{7}{4}\pi)$ ///

(2) $(\sqrt{3}, -1)$



$r^2 = x^2 + y^2 = 4$

$\tan\theta = \frac{-1}{\sqrt{3}}$

(a) $(2, -\frac{\pi}{6})$

(b) $(-2, \frac{5}{6}\pi)$ ///

3. Identify the curve by finding a Cartesian equation for the curve.

(1) $r = 2$

$r^2 = 4$

$x^2 + y^2 = 4$ Cartesian eqn

⇒ Circle with center $(0, 0)$ & radius 2. ///

(2) $r = 5\cos\theta$

$r^2 = 5r\cos\theta$

$x^2 + y^2 = 5x$ or $(x - \frac{5}{2})^2 + y^2 = \frac{25}{4}$ Cartesian eqn

⇒ Circle with center $(\frac{5}{2}, 0)$ & radius $\frac{5}{2}$. ///

4. Find a polar equation for the curve represented by the given Cartesian equation.

(1) $x = -1$

$r\cos\theta = -1$

∴ $r = -\frac{1}{\cos\theta}$ or $r = -\sec\theta$ ///

polar eqn.

(2) $x^2 + y^2 = 4y$

$r^2 = 4r\sin\theta \Rightarrow r(r - 4\sin\theta) = 0$

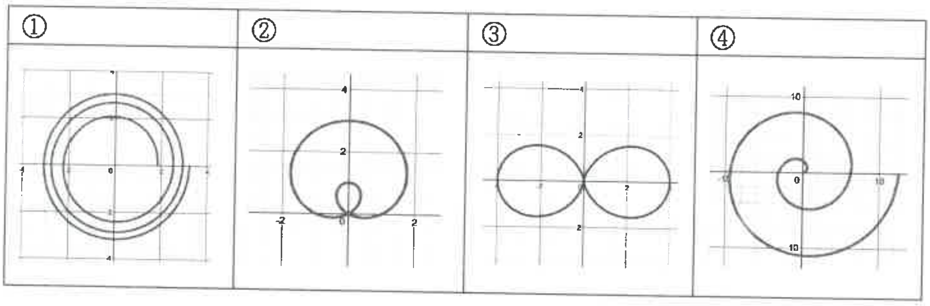
$\Rightarrow r = 0$ or $r = 4\sin\theta$

$r = 0$ is included in $r = 4\sin\theta$ when $\theta = 0$

∴ $r = 4\sin\theta$ ///

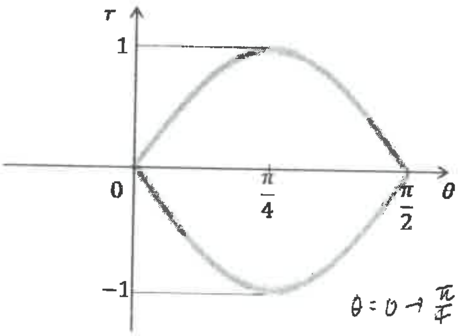
5.[기출문제] Select the right graph for the given polar equations.

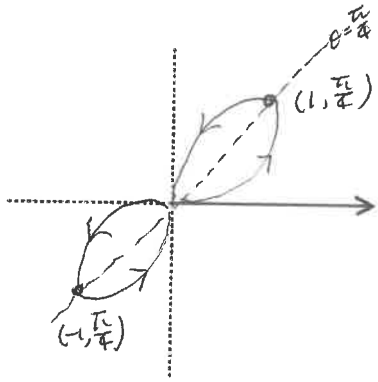
(1) $r=\theta,\ 0\le\theta\le 4\pi$ ④ (2) $r=1+2\sin\theta$ ②



6.[기출문제] (1) The polar coordinates (r,θ) of the point $(1,-1)$ in the Cartesian coordinates, where $r>0$ and $0\le\theta<2\pi$ is ($\sqrt{2}$, $\frac{7}{4}\pi$).

(2) The figure(left) shows a graph of r as a function of θ in Cartesian coordinates. Use it to sketch the corresponding polar curve. (right).





$\theta: 0 \to \frac{\pi}{4} \to \frac{\pi}{2}$

$r: 0 \to 1 \to 0$

$0 \to -1 \to 0$

3사분면

7.[기출문제] (1) Find all points whose polar coordinates lies in the third quadrant. ①②

① $\left(-1,\ \frac{\pi}{4}\right)$ ② $\left(2,\ -\frac{3\pi}{4}\right)$ ③ $\left(-3,\ \frac{5\pi}{4}\right)$ ④ $\left(4,\ -\frac{7\pi}{4}\right)$

(2) A Cartesian equation corresponding to the polar equation $r=\dfrac{4}{2\sin\theta-\cos\theta}$ is $x=$ $2y-4$

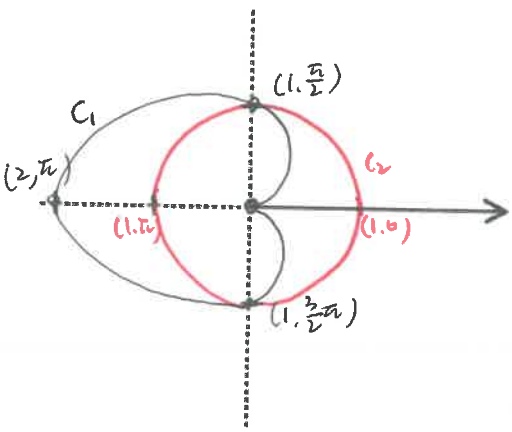
$r(2\sin\theta-\cos\theta)=4$

$2y-x=4$

§10.4 Calculus with Polar Coordinates

8.[기출문제] (1) Sketch the polar curves $r=1-\cos\theta$ and $r=1$.

C_1 C_2



(2) Find the area of the region inside of $r=1-\cos\theta$ and outside of $r=1$.

$$A=\int_{\frac{\pi}{2}}^{\frac{3}{2}\pi}\frac{1}{2}\left[(1-\cos\theta)^2-1^2\right]d\theta$$

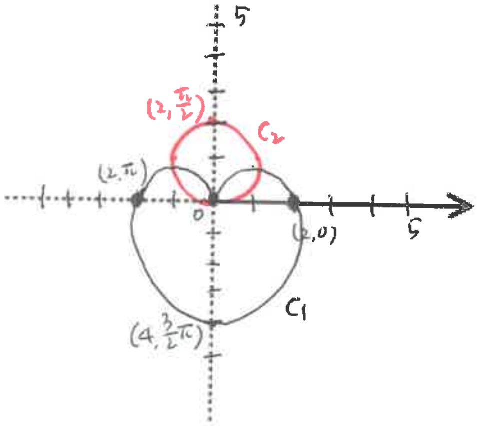
$$=2\cdot\frac{1}{2}\cdot\int_{\frac{\pi}{2}}^{\pi}\left[(1-\cos\theta)^2-1\right]d\theta$$

$$=\frac{\pi}{4}+2 \quad ///$$

9.[기출문제] Consider the polar curves

$$C_1: r=2(1-\sin\theta) \text{ and } C_2: r=2\sin\theta.$$

(1) Sketch the curves C_1 and C_2.



$r=2-2\sin\theta$

$r=2\sin\theta$

(2) Find the area of the region that lies inside both the curves C_1 and C_2.

$2(1-\sin\theta)=2\sin\theta$

$\theta=\frac{\pi}{6},\ \frac{5}{6}\pi$

$$A=2\left(\int_0^{\frac{\pi}{6}}\frac{1}{2}(2\sin\theta)^2d\theta+\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}\frac{1}{2}(2-2\sin\theta)^2d\theta\right)$$

$$=\int_0^{\frac{\pi}{6}}4\underbrace{\sin^2\theta}_{\frac{1-\cos2\theta}{2}}d\theta+\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}4\Big(1-2\sin\theta+\underbrace{\sin^2\theta}_{\frac{1-\cos2\theta}{2}}\Big)d\theta$$

$$=\left[2\theta-\sin2\theta\right]_0^{\frac{\pi}{6}}+\left[6\theta+8\cos\theta-\sin2\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$=\frac{\pi}{3}-\frac{\sqrt{3}}{2}+2\pi-4\sqrt{3}+\frac{\sqrt{3}}{2}=\frac{7\pi}{3}-4\sqrt{3} \quad ///$$

§7.8 Improper Integrals (이상적분)

10. Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

(1) $\int_0^{\infty}e^{-2x}\,dx$

$$\int_0^{\infty}e^{-2x}dx=\lim_{t\to\infty}\int_0^t e^{-2x}dx=\lim_{t\to\infty}\left[-\frac{1}{2}e^{-2x}\right]_0^t=\lim_{t\to\infty}-\frac{1}{2}\left(e^{-2t}-1\right)=\frac{1}{2}.$$

$\therefore \int_0^{\infty}e^{-2x}dx$ is convergent. ///

(2) $\displaystyle\int_e^\infty \frac{1}{x(\ln x)^2}\,dx$

$$\int_e^\infty \frac{1}{x(\ln x)^2}dx = \lim_{t\to\infty}\int_e^t \frac{1}{x(\ln x)^2}dx = \lim_{t\to\infty} -\frac{1}{\ln x}\Big]_e^t \quad (\text{치환적분 } u=\ln x)$$

$$= \lim_{t\to\infty}\left(-\frac{1}{\ln t}+1\right) = 1$$

$$\therefore \int_e^\infty \frac{1}{x(\ln x)^2}dx \text{ is convergent. ///}$$

11. Use the Comparison Theorem to determine whether the integral is convergent or divergent.

(1) $\displaystyle\int_0^\infty \frac{x}{x^3+1}\,dx = \underbrace{\int_0^1 \frac{x}{x^3+1}dx}_{\text{conv.}} + \int_1^\infty \frac{x}{x^3+1}dx$

∴ 정적분이라 수렴.

For $x>0$, $0 < \frac{x}{x^3+1} < \frac{x}{x^3} = \frac{1}{x^2}$ & $\int_1^\infty \frac{1}{x^2}dx$ is convergent

$\left(\because \lim_{t\to\infty}\int_1^t \frac{1}{x^2}dx = \lim_{t\to\infty} -\frac{1}{x}\Big]_1^t = 1\right)$

So $\int_1^\infty \frac{x}{x^3+1}dx$ is convergent by the Comparison Theorem.

$$\therefore \int_0^\infty \frac{x}{x^3+1}dx \text{ is convergent. ///}$$

(2) $\displaystyle\int_1^\infty \frac{1+\sin^2 x}{\sqrt{x}}\,dx$

For $x\ge 1$, $\frac{1+\sin^2 x}{\sqrt{x}} \ge \frac{1}{\sqrt{x}} > 0$ & $\int_1^\infty \frac{1}{\sqrt{x}}dx$ is divergent by Thm 2 with $p=\frac{1}{2}<1$.

$\left(\text{or } \lim_{t\to\infty}\int_1^t \frac{1}{\sqrt{x}}dx = \lim_{t\to\infty} 2\sqrt{x}\Big]_1^t = \lim_{t\to\infty} 2(\sqrt{t}-1) = \infty\right)$

So $\int_1^\infty \frac{1+\sin^2 x}{\sqrt{x}}dx$ is divergent by the Comparison Theorem. ///

(3) $\displaystyle\int_0^\infty \frac{\arctan x}{2+e^x}\,dx$

For $x\ge 0$, $\arctan x < \frac{\pi}{2} < 2$ so $0 \le \frac{\arctan x}{2+e^x} < \frac{2}{2+e^x} < \frac{2}{e^x} = 2e^{-x}$

Now $\int_0^\infty 2e^{-x}dx = \lim_{t\to\infty}\int_0^t 2e^{-x}dx = \lim_{t\to\infty} -2e^{-x}\Big]_0^t = \lim_{t\to\infty}\left(-\frac{2}{e^t}+2\right) = 2$ conv.

By Comparison Theorem, $\int_0^\infty \frac{\arctan x}{2+e^x}dx$ is convergent ///

§11.1 Sequences (수열)

12. 단조수열정리(Monotonic Sequence Theorem)를 쓰시오.

Every bounded, monotonic sequence is convergent.

수열이 단조이고 유계이면, 그 수열은 수렴한다.

§11.2 Series (급수)

13. 발산판정법(Test for Divergence)을 쓰시오.

If $\lim_{n\to\infty} a_n$ does not exist or $\lim_{n\to\infty} a_n \neq 0$, then Σa_n is divergent.

$\lim_{n\to\infty} a_n$이 존재하지 않거나 $\lim_{n\to\infty} a_n \neq 0$ 이면 급수 Σa_n은 발산한다.

14. Let $a_n = \dfrac{2n}{3n+1}$.

(1) Determine whether $\{a_n\}$ is convergent.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \dfrac{2n}{3n+1} = \dfrac{2}{3}$ $\therefore$ $\left\{\dfrac{2n}{3n+1}\right\}$ is convergent ///

(2) Determine whether $\sum_{n=1}^{\infty} a_n$ is convergent.

Since $\lim_{n\to\infty} a_n = \frac{2}{3} \neq 0$, $\Sigma \dfrac{2n}{3n+1}$ is divergent by Test for Divergence. ///

15. Find the sum of the series $\sum_{n=1}^{\infty} a_n$ whose partial sum $s_n = \dfrac{n^2-1}{4n^2+1}$.

Since $\lim_{n\to\infty} s_n = \lim_{n\to\infty} \dfrac{n^2-1}{4n^2+1} = \dfrac{1}{4}$, $\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n = \dfrac{1}{4}$ ///

16. Find the values of c if $\sum_{n=2}^{\infty} (1+c)^{-n} = 2$.

geometric series with $a = (1+c)^{-2}$ & $r = (1+c)^{-1}$.

So the series converges when $|(1+c)^{-1}| < 1 \iff |1+c| > 1 \iff 1+c > 1$ or $1+c < -1$

$\iff c > 0$ or $c < -2$.

$\sum_{n=2}^{\infty} (1+c)^{-n} = \dfrac{a}{1-r} = \dfrac{(1+c)^{-2}}{1-(1+c)^{-1}} = 2 \iff 2c^2+2c-1=0$

$\iff c = \dfrac{-1 \pm \sqrt{3}}{2}$

$\therefore$ $c = \dfrac{-1+\sqrt{3}}{2}$ ///

수고하셨습니다